

Foundation (Jalapeno)

Q1. Solve for x in $|x| = 5$.

- (A) $x = 5$
- (B) $x = -5$
- (C) $x = 5$ or $x = -5$
- (D) $x = 0$
- (E) $x = 10$

Q2. Solve for x in $|2x| = 8$.

- (A) $x = 2$ or $x = -2$
- (B) $x = 4$ or $x = -4$
- (C) $x = 1$ or $x = -1$
- (D) $x = 8$ or $x = -8$
- (E) $x = 16$ or $x = -16$

Q3. A stock's value is $|x - 10|$. If the value is 5, what is x ?

- (A) $x = 5$
- (B) $x = 15$
- (C) $x = 5$ or $x = 15$
- (D) $x = 10$
- (E) $x = 20$

Q4. Solve for x in $|x + 2| = 3$.

- (A) $x = 1$
- (B) $x = -1$
- (C) $x = 1$ or $x = -5$
- (D) $x = -1$ or $x = 5$
- (E) $x = 0$

Q5. A company's tax is $|0.2x - 100|$. If the tax is 20, what is x ?

- (A) $x = 400$
- (B) $x = 600$
- (C) $x = 400$ or $x = 600$
- (D) $x = 500$
- (E) $x = 1000$

Q6. Solve for x in $|3x - 1| = 11$.

- (A) $x = 4$
- (B) $x = -4$
- (C) $x = 4$ or $x = -4$
- (D) $x = 2$ or $x = -2$
- (E) $x = 12$ or $x = -12$

Q7. A savings account's interest is $|0.05x|$. If the interest is 2.50, what is x ?

- (A) $x = 25$
- (B) $x = 50$
- (C) $x = 25$ or $x = -50$
- (D) $x = 50$ or $x = -25$
- (E) $x = 100$ or $x = -100$

Q8. Solve for x in $|x - 5| = |x + 3|$.

- (A) $x = 1$
- (B) $x = -1$
- (C) $x = 1$ or $x = -1$
- (D) $x = 0$
- (E) $x = 2$

Open Ended Questions (FRQ)

Q9. Solve for x in $|2x + 1| = |x - 4|$. Show all steps and check for extraneous solutions.

Q10. A business's profit is $|0.1x - 50|$. If the profit is 10, what is x ? Show all steps and check for extraneous solutions.

Chef's Tips

- Check for extraneous solutions
- Use the definition of absolute value to solve equations
- Consider both positive and negative solutions